

EC2 Auto Start/Stop Deployment Portfolio Project

1. Objective

To automate starting and stopping of an EC2 instance using **AWS Lambda** and **Amazon EventBridge** schedules, ensuring cost-efficiency and automated operations.

2. Prerequisites

- AWS Account with permissions to create:
 - Lambda functions
 - IAM roles
 - EventBridge schedules
 - EC2 access
 - EC2 instance already running
 - IAM Role: `lambda-ec2-access` with `ec2:StartInstances` and `ec2:StopInstances` permissions
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3. Create a Lambda Function

What is lambda function

Lambda function is a stateless, event-driven function that can run in response to triggers such as HTTP requests, database changes, or scheduled events, and scales automatically with demand.

Navigate to **AWS Console** → **Lambda** → **Functions** → **Create Function**

1. Configure function:

- **Function name:** `lambda-instance`
- **Runtime:** `Python 3.x`
- Click **Change default execution role** → **Use another role** → **Role:** `lambda-ec2-access`

2. Click **Create function**

4. Lambda Function Script

```
import boto3

ec2=boto3.client('ec2')

INSTANCE_ID='i-063c4c7f558a84e08'

def lambda_handler(event,context):

    action=event.get('action')

    if action=='start':

        ec2.start_instances(InstanceIds=[INSTANCE_ID])

        return "EC2 Start ho gaya"

    elif action=='stop':

        ec2.stop_instances(InstanceIds=[INSTANCE_ID])

        return "EC2 Stop ho gaya"

    else:

        return "Invalid action"
```

Deployment: Click the **Deploy** icon to save changes.

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5. Create EventBridge Schedule – EC2 Start

What is Amazon EventBridge

Amazon EventBridge is a serverless event bus service from AWS that allows you to connect applications using events.

1. Navigate to **Amazon EventBridge** → **Schedules** → **Create Schedule**

2. Configure schedule:

- **Schedule name:** `start-ec2-Schedules`
- **Description:** Optional – `Start EC2 instance`
- **Time zone:** `(UTC+05:30) Asia/Kolkata`
- **Schedule type:** `Cron-based schedule`
- **Cron expression:**

`0 10 * * ? *`

Minutes Hours Day of Month Month Day of Week Year

`0   10   *   *   ?   *`

- **Flexible time window:** Off

3. Click **Next** → **Select target:** AWS Lambda → **Next**
4. **Lambda function:** `lambda-instance`
5. **Payload:**

```
{  
  "action": "start"  
}
```

6. Click **Next** → **Next** → **Create Schedule**

6. Create EventBridge Schedule – EC2 Stop

1. Navigate to **Amazon EventBridge** → **Schedules** → **Create Schedule**
2. Configure schedule:

- **Schedule name:** `stop-ec2-Schedules`
- **Description:** Optional – `Stop EC2 instance`
- **Time zone:** `(UTC+05:30) Asia/Kolkata`
- **Schedule type:** `Cron-based schedule`
- **Cron expression:**

`0 22 * * ? *`

Minutes	Hours	Day of Month	Month	Day of Week	Year
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<code>0</code>	<code>22</code>	<code>*</code>	<code>*</code>	<code>?</code>	<code>*</code>
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- **Flexible time window:** Off
- 3. Click **Next** → **Select target:** AWS Lambda → **Next**
- 4. **Lambda function:** `lambda-instance`
- 5. **Payload:**

```
{  
  "action": "stop"  
}
```

- 6. Click **Next** → **Next** → **Create Schedule**
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7. Test & Verify

1. **Manual Lambda Test:**
 - Event { `"action": "start"` } → EC2 should start
 - Event { `"action": "stop"` } → EC2 should stop
 2. **EventBridge Schedule Test:**
 - Click **Run** on schedule → Lambda triggers → EC2 starts/stops
 3. **Check EC2 Status:**
 - AWS Console → EC2 → Instances → Verify running/stopped status
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8. Final Result

- EC2 instance **automatically starts at 10:00 AM**
- EC2 instance **automatically stops at 10:00 PM**

- Lambda function handles the action via **EventBridge schedule**
 - **Manual start/stop** also works without any issue
 - **CloudWatch logs** track all Lambda executions
 - Cost-efficient automated solution
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 Conclusion:

This setup ensures that your EC2 instance is **fully automated**, **cost-effective**, and **safe**, using AWS Lambda and EventBridge schedules.